SOFTWARE ENGINEERING LAB · PROJECT-BASED LEARNING

Smart Classroom Electricity Management System

Vacancy detection, AC-state inference and appliance control

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Software Engineering Lab (24CS01PR0502) · 2026-27

2
modules built

78
test cases

45
Jira work items

Why classrooms waste electricity

1 Appliances stay ON

Lights, fans and ACs keep running after a lecture ends. Manual switching and fixed timers cannot follow irregular schedules.

2 Sensors miss the picture

PIR sensors cannot see whether fans or lights are on, and miss people who sit still.

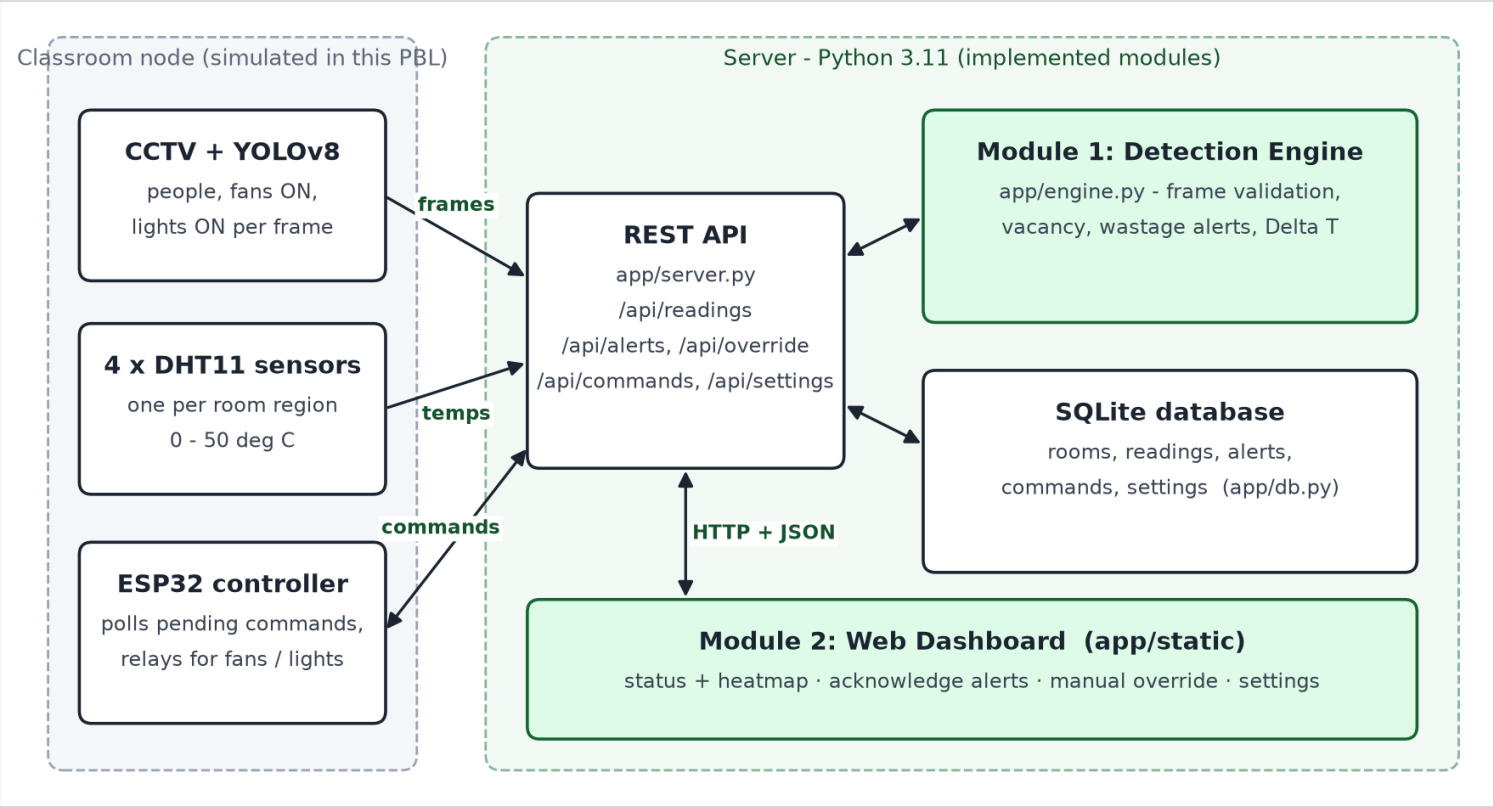
3 AC state is invisible

One temperature sensor cannot tell whether the AC is still cooling an empty room.

Proposed system

CCTV + YOLOv8 counts people, fans and lights; four DHT11 sensors read four room regions; an ESP32 switches relays; a web dashboard shows alerts and lets an operator act.

What this PBL builds



Implemented

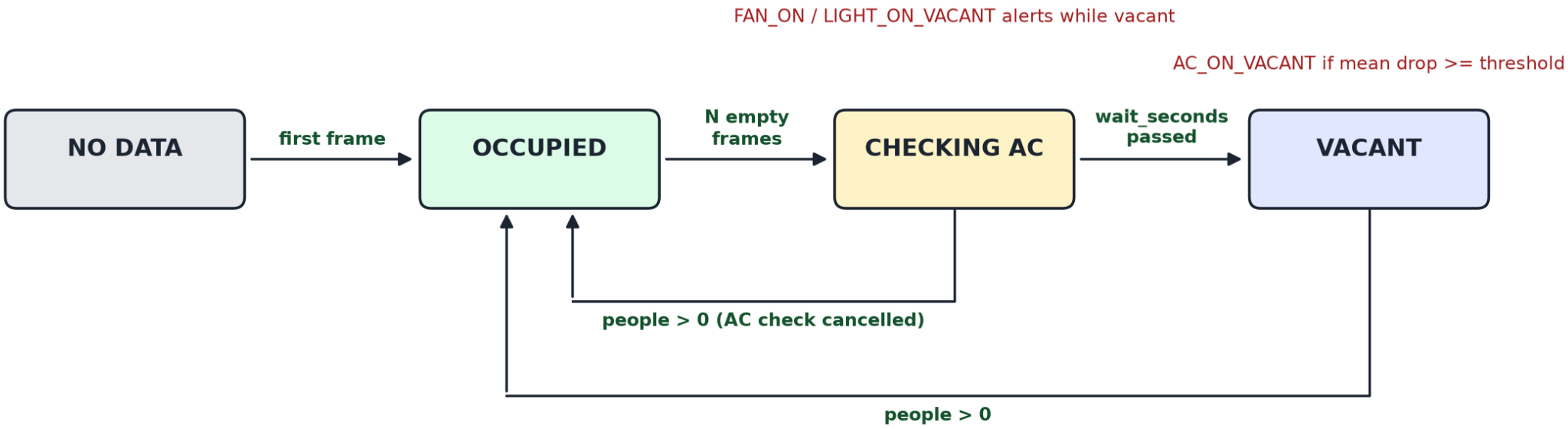
Module 1 - detection engine
Module 2 - REST API, SQLite database and web dashboard

Simulated

CCTV + YOLOv8, four DHT11 sensors and the ESP32 are replaced by app/simulator.py, which sends the same data

Python 3.11 standard library · SQLite · HTML, CSS, JavaScript

Module 1 · Vacancy and AC-state detection engine



Vacancy

Room is vacant only after N consecutive frames with no people (default 10). Any person resets the count.

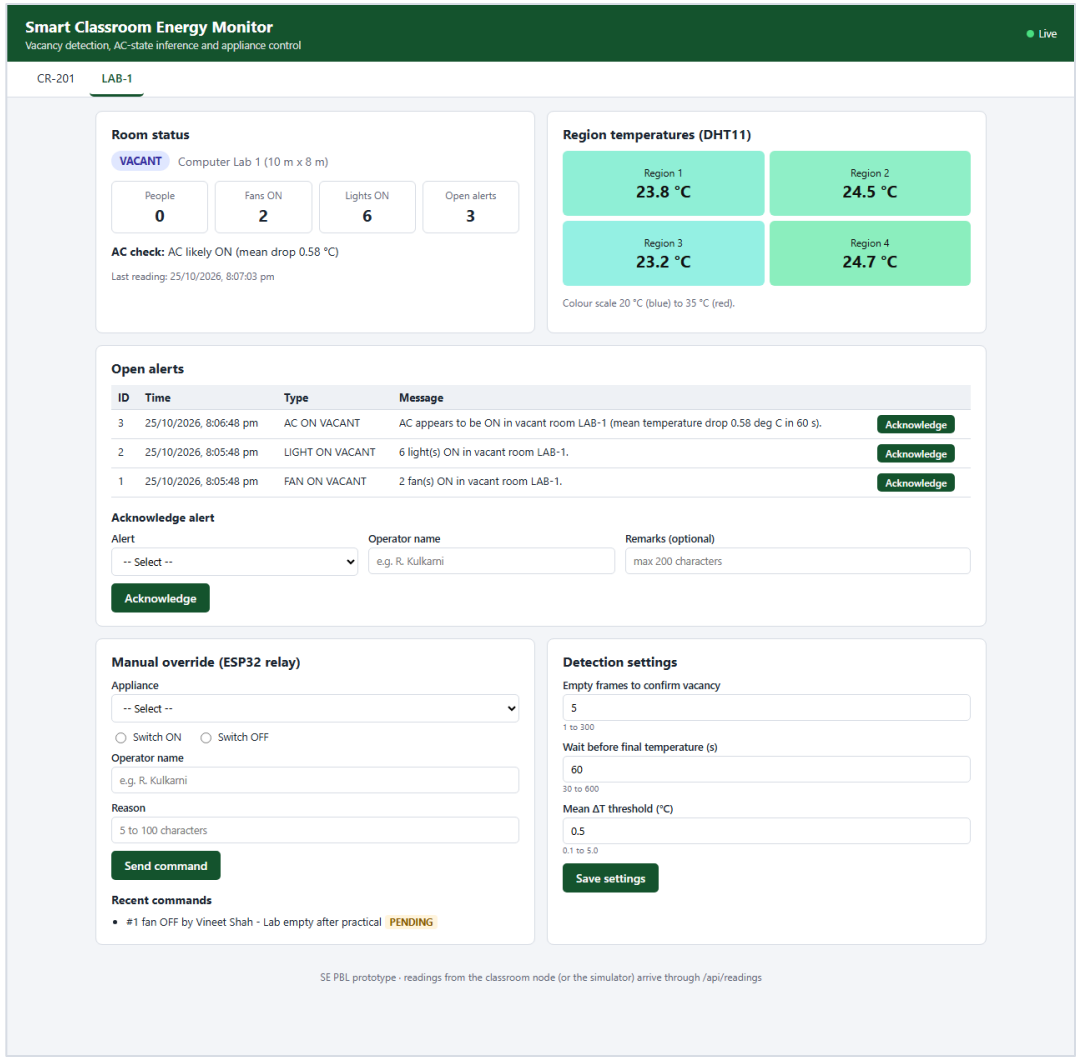
Wastage alerts

FAN_ON_VACANT and LIGHT_ON_VACANT are raised while the room is vacant, once per occurrence.

AC inference

After wait_seconds, mean temperature drop over the 4 regions >= threshold means the AC is still ON.

Module 2 · Monitoring dashboard and control



1

Live room status

state badge, people, fans and lights ON, AC check, 2x2 heatmap

2

Alerts

open alerts listed newest first; operator acknowledges with name and remarks

3

Manual override

fan or light ON/OFF queued for the ESP32, marked done when executed

4

Settings

vacancy frames, wait time and threshold, validated and stored in SQLite

5

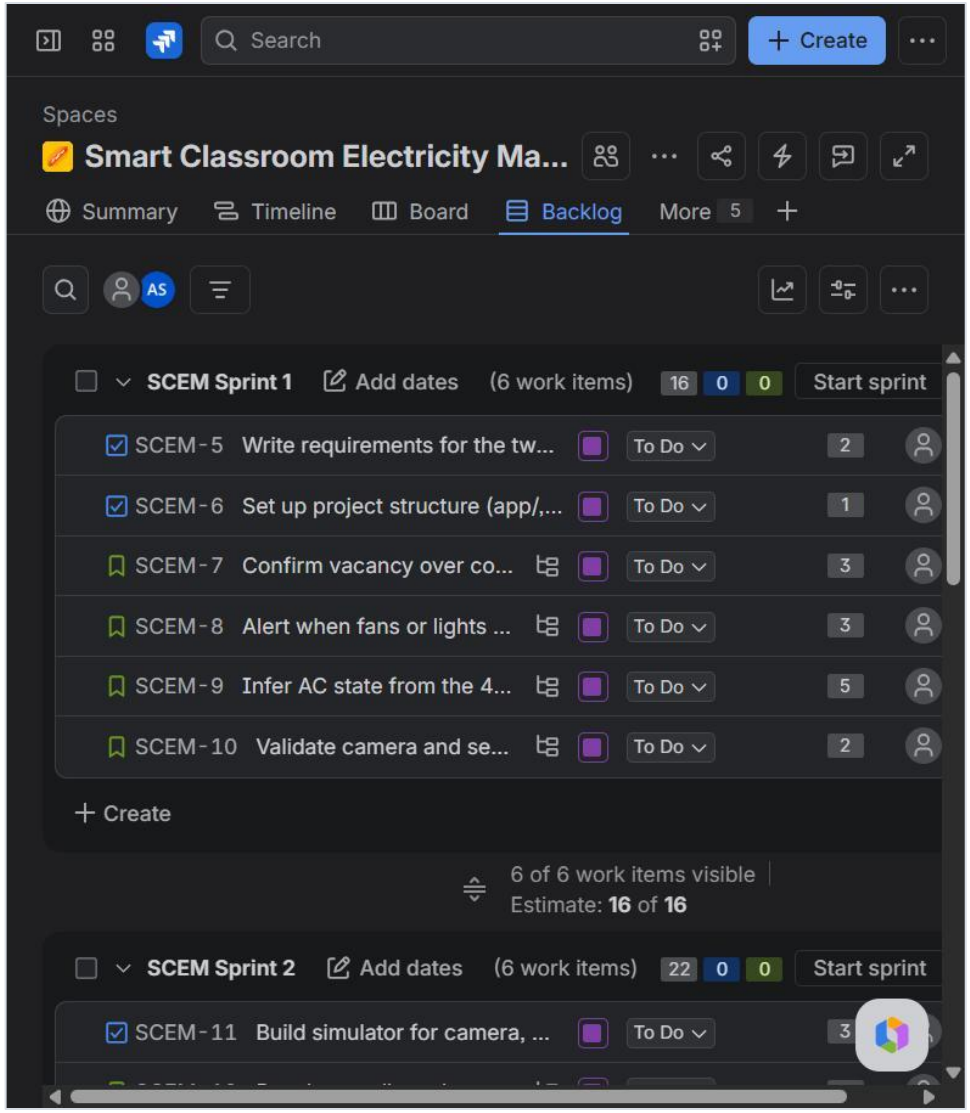
REST API

/api/readings, /alerts, /override, /commands, /settings

Managed in Jira · space SCEM

4 epics	**9** user stories	**10** tasks
18 subtasks	**4** bugs	**3** sprints

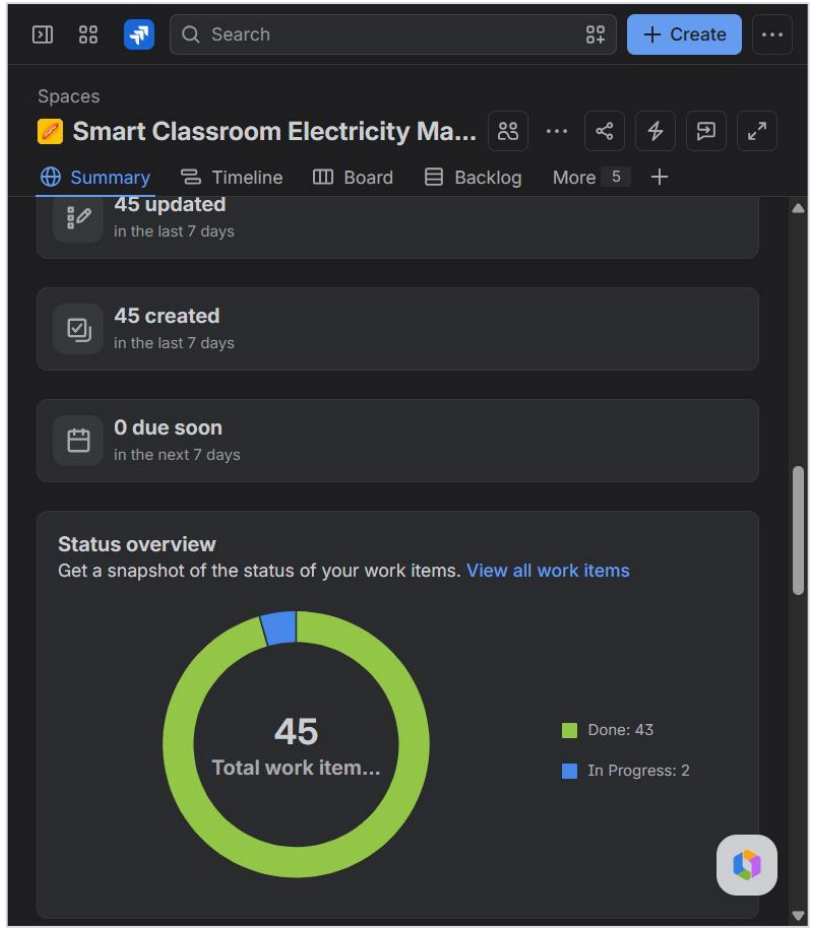
- Epic -> Story / Task / Bug -> Subtask
- Backlog imported from jira/jira_import.csv
- Workflow: To Do -> In Progress -> Done
- Each bug carries steps, severity, priority and a fix + retest comment



Three sprints, tracked to Done

Sprint	Goal	Items	Points	State
SCEM Sprint 1 **25-10 19:30-19:50**	Detection engine (Module 1) and requirements	6	16	Completed
SCEM Sprint 2 **25-10 19:50-20:00**	Dashboard, API, database and simulator (Module 2)	6	22	Completed
SCEM Sprint 3 **25-10 20:00 - 01-11**	Testing, bug fixing and documentation	11	16	Active

Sprint 3 holds the testing tasks and the four bugs; the report and presentation task stays In Progress until submission.



78 automated test cases

32	24	7	15
Unit tests engine	API tests server + DB	Integration end to end	UI tests in Edge

✓ **Functional**	every requirement checked through its output
✓ **Equivalence partitioning**	valid and invalid classes for each input
✓ **Boundary value analysis**	min-1, min, max, max+1 for every limit
✓ **Validation**	every API and form error message
✓ **Unit and integration**	engine alone, then engine + API + DB + ESP32 loop

Format: Test Case ID · Module · Test Scenario · Test Data · Expected Result · Actual Result · Status

Run 1 found 4 real bugs

73 / 5

Run 1: passed / failed

→

78 / 0

Run 2 (retest): passed / failed

- Each failure was analysed, logged as a Bug in Jira, fixed in code and retested.
- Fix and retest comments are on each bug (SCEM-24 to SCEM-27).

BUG-01 · SCEM-24 — Minor / Medium

Room with no readings shown as OCCUPIED

BUG-02 · SCEM-25 — Major / High

False AC alert at 0.499 deg C drop (rounding)

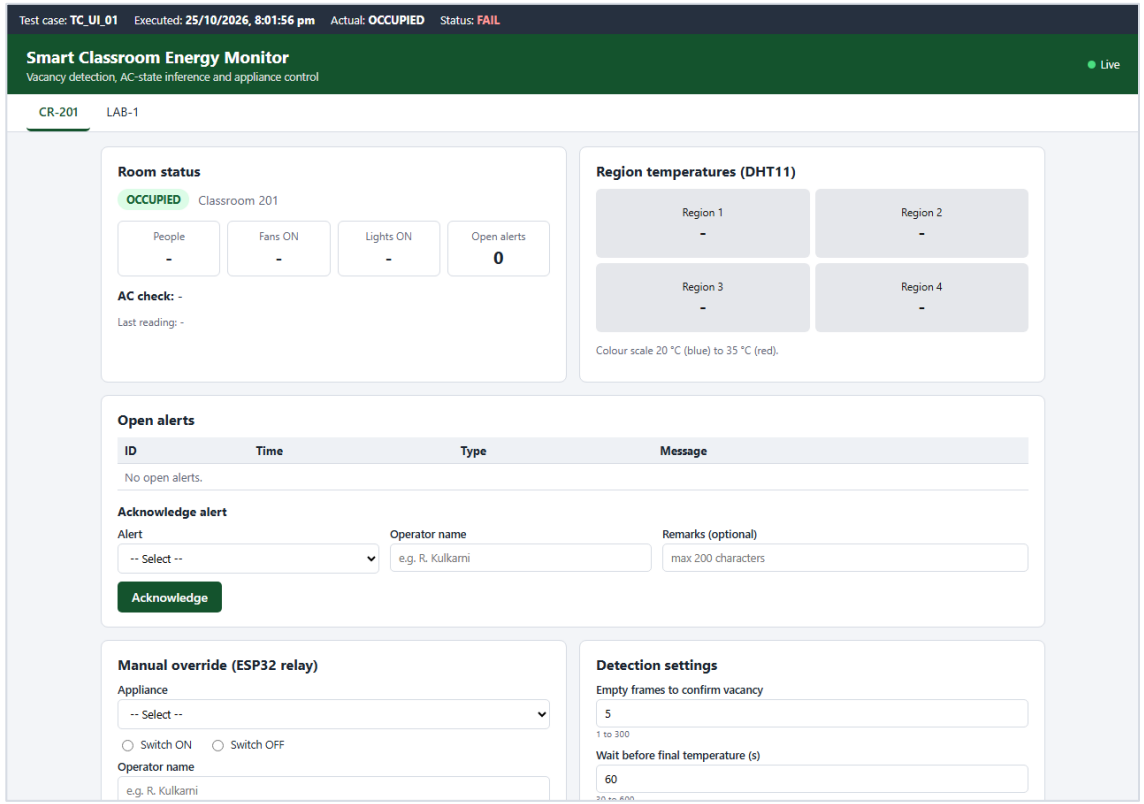
BUG-03 · SCEM-26 — Minor / Low

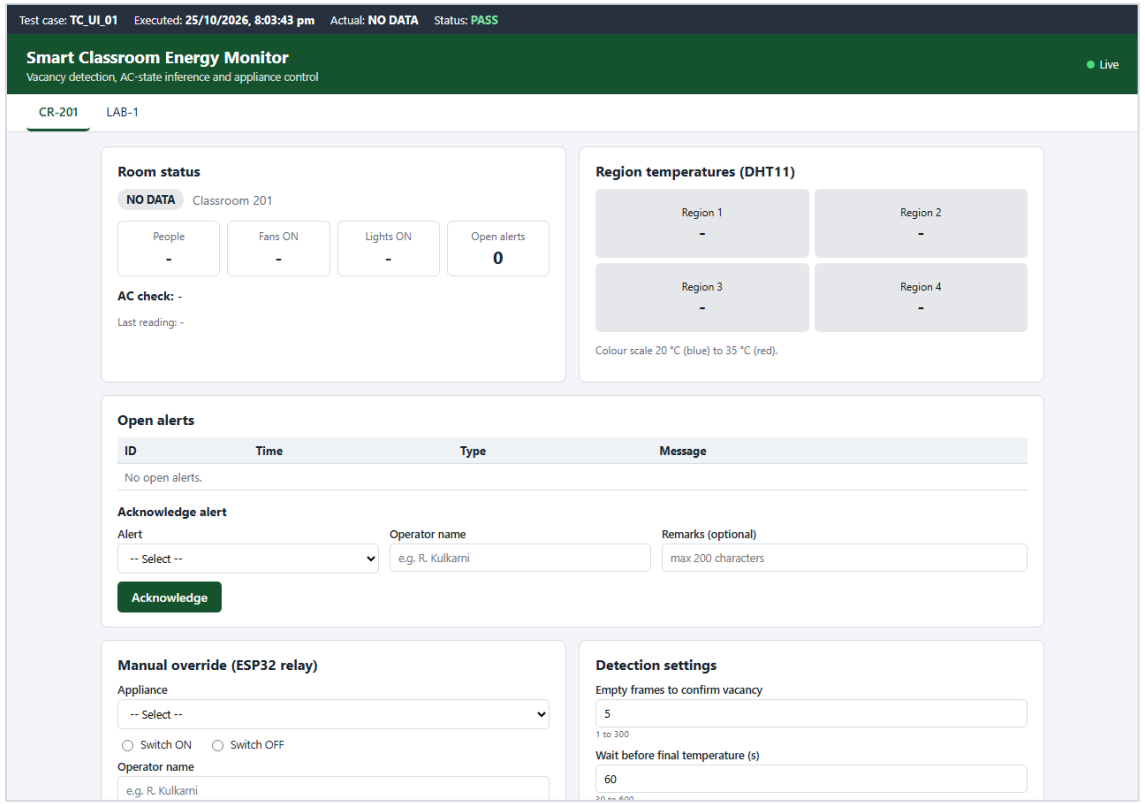
Missing room_id gave 404 'Room None'

BUG-04 · SCEM-27 — Trivial / Low

Settings errors showed internal field names

Bug fix and retest evidence · BUG-01





Before fix: OCCUPIED, TC_UI_01 FAIL

After fix: NO DATA, TC_UI_01 PASS

Same seeded data, same test case, captured from the real dashboard in Microsoft Edge before and after the fix

Live demonstration

1

Start the server

```
python -m app.server --port 8090
```

2

Run the classroom simulator

```
python -m app.simulator --room LAB-1 --scenario ac_left_on
```

3

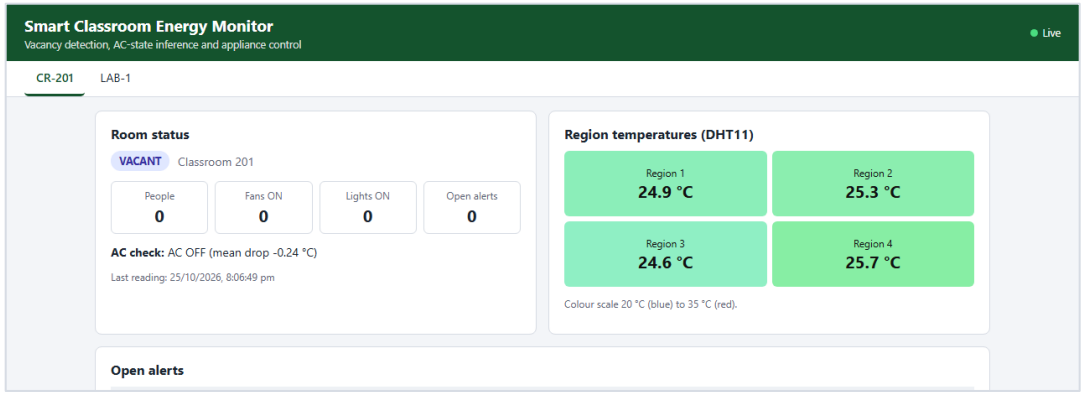
Watch and act

alerts appear; acknowledge, override a fan, change a setting

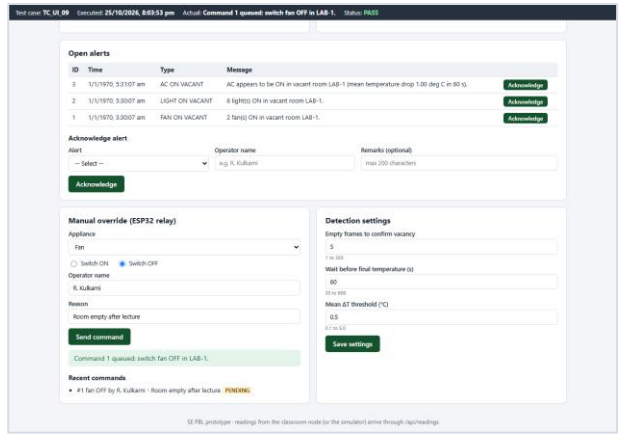
4

Run all tests

```
python -m tests.run_tests
```



CR-201 after everything was switched off: AC inferred OFF (room warmed by 0.24 deg C)



Conclusion and next steps

What was achieved

- Two working modules with database, validation and integration
- Jira project: 45 work items, 3 sprints, bugs tracked to closure
- 78 test cases; 4 bugs found, fixed and retested; 78 / 78 pass

Next steps

- Connect the real CCTV + YOLOv8, DHT11 sensors and ESP32
- MQTT between classroom node and server
- Daily energy-wastage report and mobile alerts

Thank you · Questions?